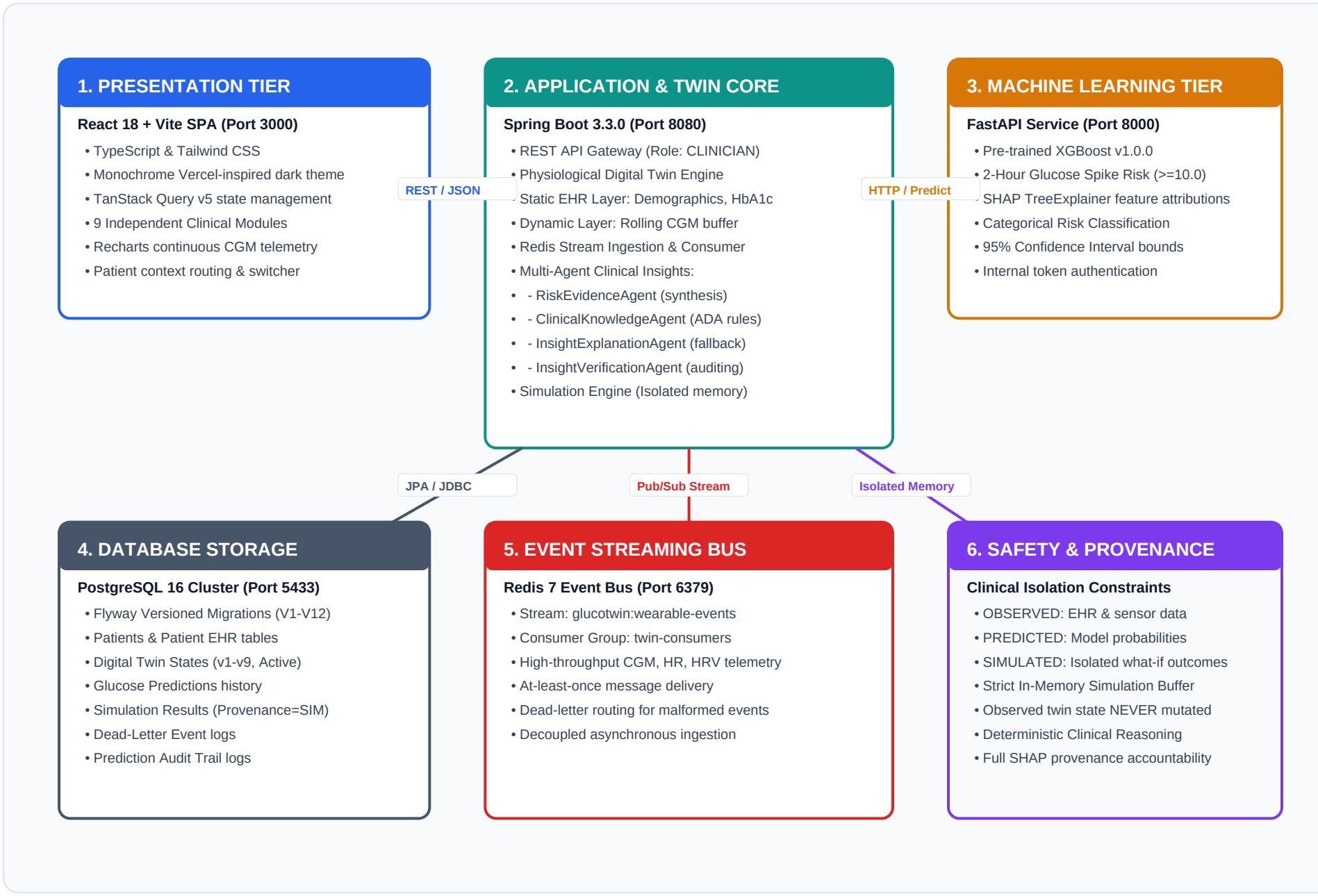
GlucoTwin AI — End-to-End System Architecture

Clinical Decision Support & Physiological Digital Twin Platform for Type 2 Diabetes


1. PRESENTATION TIER
React 18 + Vite SPA (Port 3000)
• TypeScript & Tailwind CSS
• Monochrome Vercel-inspired dark theme
• TanStack Query v5 state management
• 9 Independent Clinical Modules
• Recharts continuous CGM telemetry
• Patient context routing & switcher
REST / JSON
2. APPLICATION & TWIN CORE
Spring Boot 3.3.0 (Port 8080)
• REST API Gateway (Role: CLINICIAN)
• Physiological Digital Twin Engine
• Static EHR Layer: Demographics, HbA1c
• Dynamic Layer: Rolling CGM buffer
• Redis Stream Ingestion & Consumer
• Multi-Agent Clinical Insights:
• - RiskEvidenceAgent (synthesis)
• - ClinicalKnowledgeAgent (ADA rules)
• - InsightExplanationAgent (fallback)
• - InsightVerificationAgent (auditing)
• Simulation Engine (Isolated memory)
HTTP / Predict
3. MACHINE LEARNING TIER
FastAPI Service (Port 8000)
• Pre-trained XGBoost v1.0.0
• 2-Hour Glucose Spike Risk (>=10.0)
• SHAP TreeExplainer feature attributions
• Categorical Risk Classification
• 95% Confidence Interval bounds
• Internal token authentication
JPA / JDBC
Pub/Sub Stream
Isolated Memory
4. DATABASE STORAGE
PostgreSQL 16 Cluster (Port 5433)
• Flyway Versioned Migrations (V1-V12)
• Patients & Patient EHR tables
• Digital Twin States (v1-v9, Active)
• Glucose Predictions history
• Simulation Results (Provenance=SIM)
• Dead-Letter Event logs
• Prediction Audit Trail logs
5. EVENT STREAMING BUS
Redis 7 Event Bus (Port 6379)
• Stream: glucotwin:wearable-events
• Consumer Group: twin-consumers
• High-throughput CGM, HR, HRV telemetry
• At-least-once message delivery
• Dead-letter routing for malformed events
• Decoupled asynchronous ingestion
6. SAFETY & PROVENANCE
Clinical Isolation Constraints
• OBSERVED: EHR & sensor data
• PREDICTED: Model probabilities
• SIMULATED: Isolated what-if outcomes
• Strict In-Memory Simulation Buffer
• Observed twin state NEVER mutated
• Deterministic Clinical Reasoning
• Full SHAP provenance accountability